Can It Ford? Where a Standard Flood-Safety Rule Fails Against Simulated Physics

Josie Cerrell | jcerrell29@cmc.edu

Mentors: Cheng-Hsi Hsiao, Dr. Hassan Iqbal, Dr. Krishna Kumar

Integrated Sciences, Claremont McKenna College | Texas Advanced Computing Center, The University of Texas at Austin

TEXAS ADVANCED COMPUTING CENTER

KEY FINDING **Correcting our own encoding of the standard criterion reclassified 23 of 70 scenario cells. All 23 moved toward NO-FORD. Zero were loosened.**

Introduction

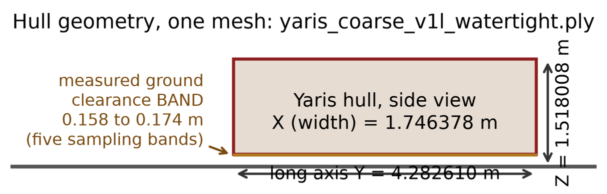
Flooded roads kill drivers every year; an autonomous vehicle needs a rule for when to stop. The 23 reclassified cells decompose: 11 come from restoring the depth cap with the vehicle class held fixed, and all 11 are still or slow water 0.6 to 1.0 m deep. The other 12 come from evaluating a 1100 kg car against its own published class rather than the Large 4WD limits. Both corrections tighten.

Research Goal

The field decides fordability with a hazard product: depth times velocity against a threshold. It is cheap and it is published. Checked against a physics simulation of the same scenario, does that rule agree?

Methods

An abstraction ladder on identical inputs. L0: NWS Turn Around Don't Drown, 0.15 m. L1a: AR&R depth and DxV caps applied jointly per class; its 3.0 m/s velocity cap never binds on this grid. L1b: Kramer total head, formula reconstructed here. L2: coupled material-point simulation, 20 runs.



Geometry and physics gates, measured this session

gate	measured value	verdict
mesh containment of solid particles	100.00 pct of 2000 sampled	PASS
fill_ratio vs 0.95 to 1.10 band	1.0024 at g64; 0.9941 to 1.0262 (17 runs)	PASS
water surface open_edges	0	PASS
water volume error	+1.68% raw, +1.65% smoothed	PASS
oob_particle_frames	0	PASS
determinism_identical, all 17 runs	True	PASS
realized rho vs the 100 to 300 band	310 / 453 / 658 kg/m3 at grid 64	FAIL
water particle passthrough	7.3 to 15.9 pct; worst at v = 3.0 m/s	NOTED

Fig 1. Hull geometry and the eight verification gates. Six PASS, one FAIL, one NOTED. Realized density 302.6 to 663.6 kg/m3 against this project's own 100 to 300 band is shown as a FAIL. Script: analysis/make_poster_figures.py from rollout.npz and hero_fixed.log.

Figures and Results

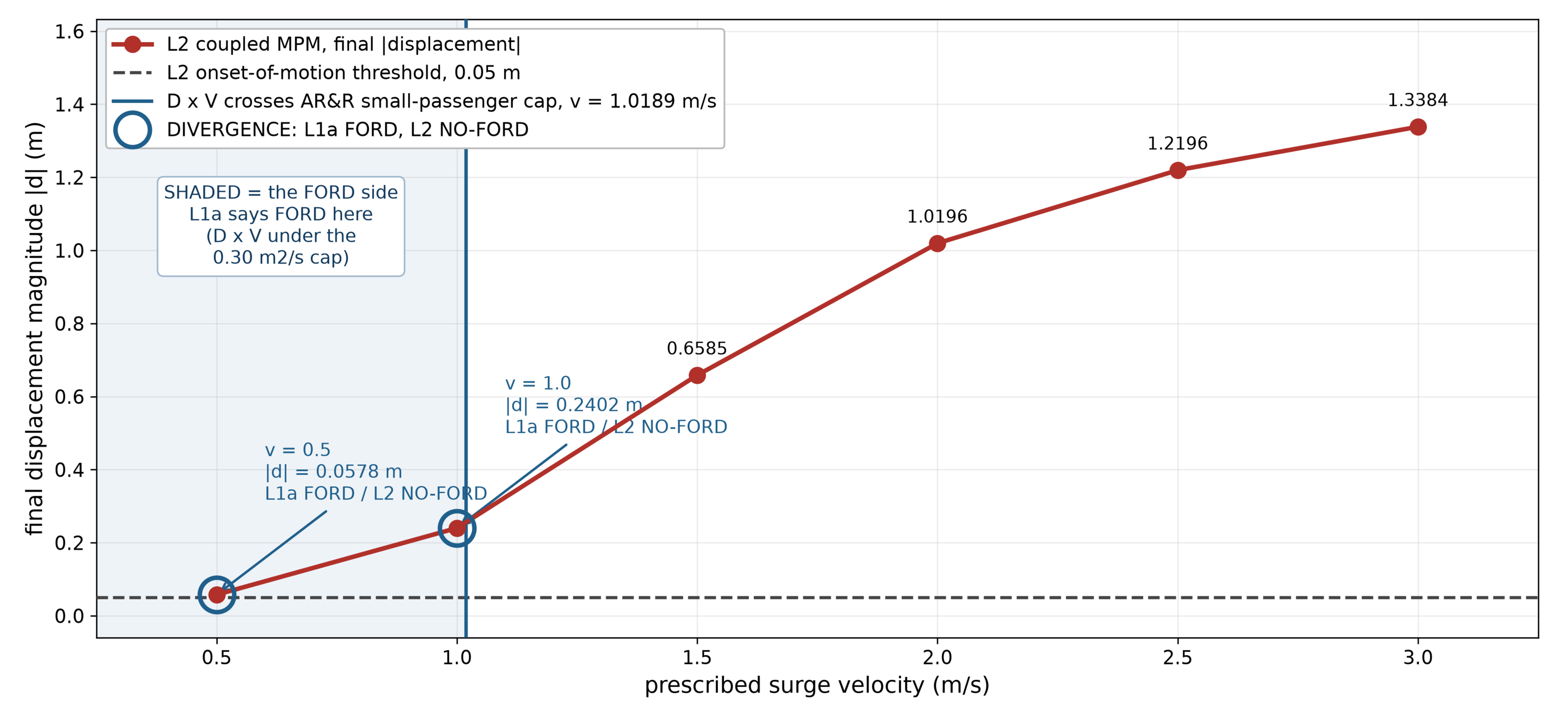


Fig 2. Final displacement against surge velocity at fixed realized depth 0.2944 m, grid 64, one hull at 1100 kg, all runs deterministic. Vertical rule marks v = 1.0189 m/s, where DxV crosses the AR&R small-passenger 0.30 m2/s cap. At v = 0.5 and v = 1.0 the criterion says FORD and the simulation says NO-FORD. The criterion is varied along the very axis it is built on and still returns the wrong verdict at two of six points. Those two points are not equally strong. At v = 1.0 the vehicle moves 0.2402 m, which is 1.6 grid cells and survives any onset threshold up to 0.2249 m. At v = 0.5 it moves 0.0578 m, clearing the 0.05 m threshold by 0.0078 m and amounting to 0.8 of one particle spacing, so that point is reported as marginal. No zero-velocity control run exists to establish the numerical floor. Script: analysis/make_poster_figures.py from data/all_runs_inventory.csv.

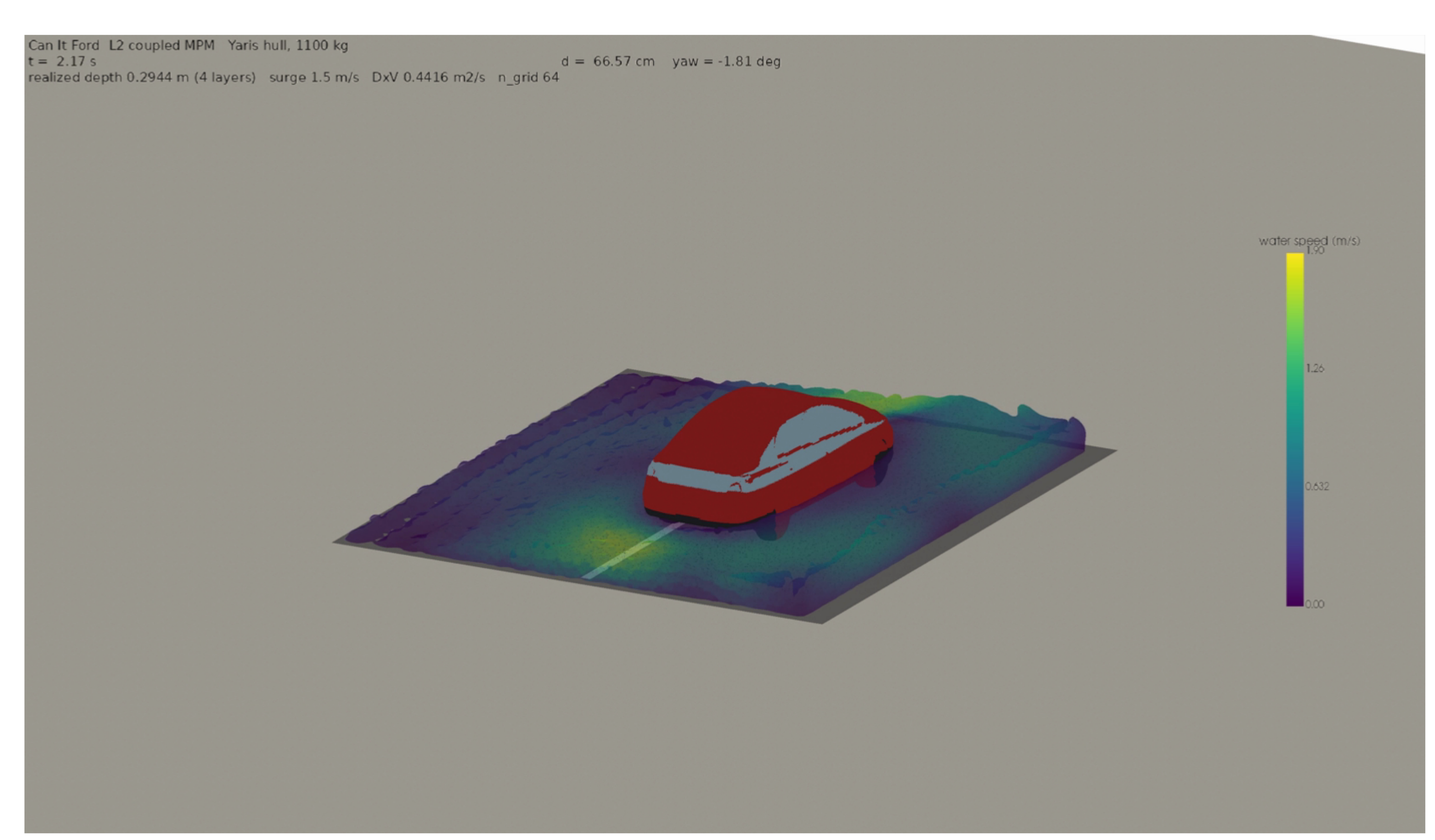


Fig 3. Frame 65 of 90, coupled standing-water run at 1100 kg, rendered after a shadow-pass fix. Water surface closes: open_edges = 0, volume error +1.68 pct. Script: renders/yaris_render_s1/render_pv.py; frame selected by analysis/make_poster_figures.py from metrics.csv.

Why The Rule Fails

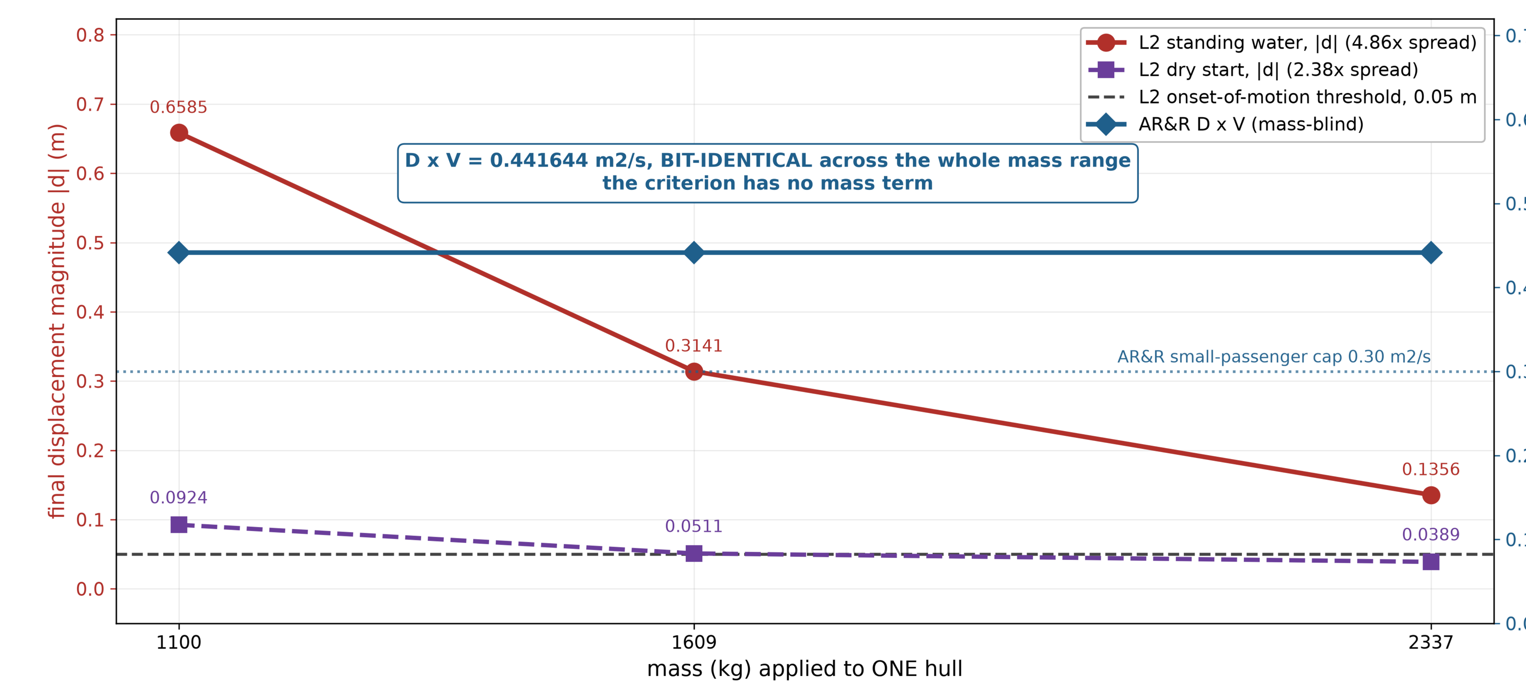


Fig 4. Final displacement against vehicle mass, standing water and dry start, grid 64. Nominal DxV is bit-identical at 0.441644 m2/s across a 2.1x mass range while simulated displacement changes 4.86x at this grid (1.87x at grid 48, 3.00x at grid 96). Three masses on one hull: a mass sensitivity study, not three vehicles. Script: analysis/make_poster_figures.py from data/all_runs_inventory.csv (standing) and gates_results.json (dry start).

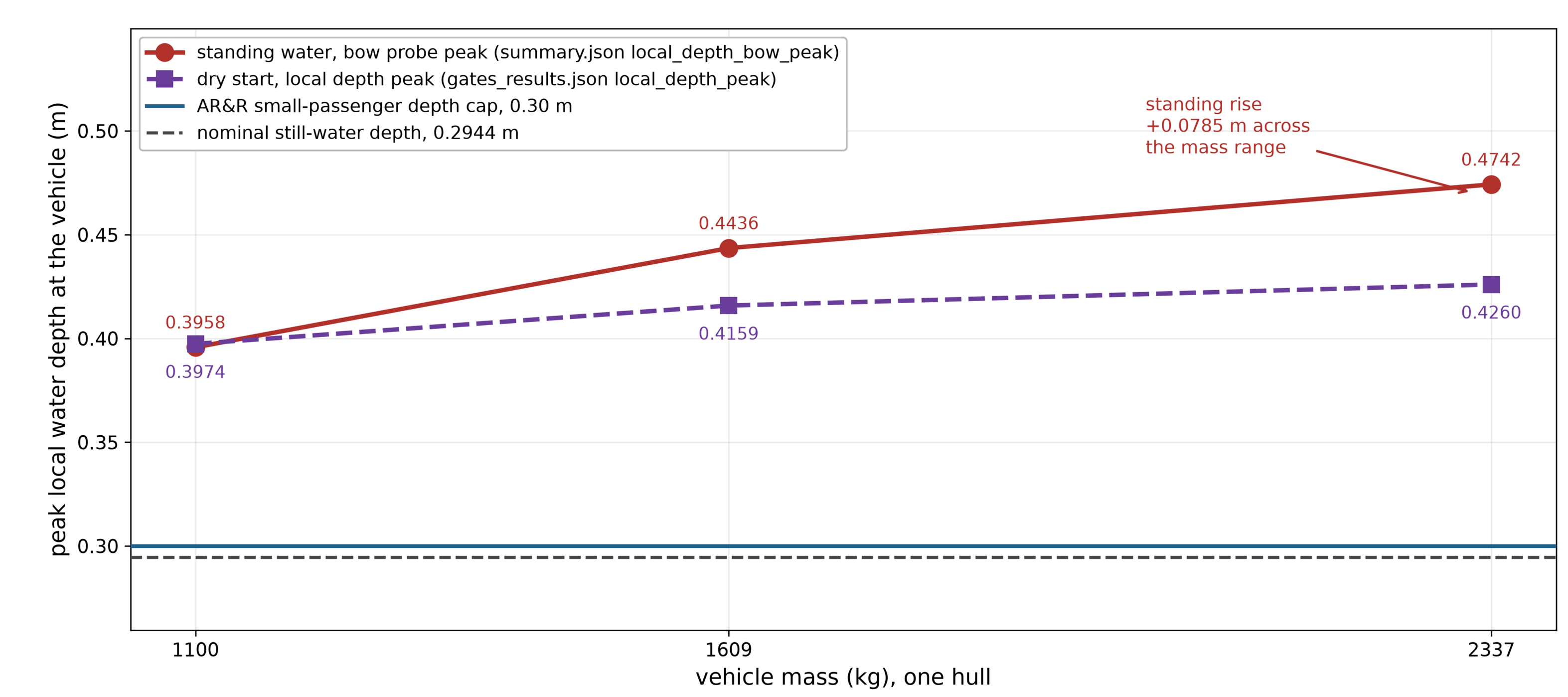


Fig 5. Peak local water depth at the vehicle against mass. Realized still-water depth 0.2944 m sits under the AR&R 0.30 m cap, yet local depth at the vehicle exceeds that cap in 19 of the 20 runs, and the exceedance grows with mass. The one exception is the slowest run, v = 0.5 m/s, at 0.295361 m. The criterion uses road depth; the vehicle sits in its own bow wave. Script: analysis/make_poster_figures.py.

Conclusion

Against the AR&R small-passenger limit set, the hazard product fails three separable ways: the wrong verdict at 2 of 6 points on its own velocity axis, blindness across a 2.1x mass range, and measuring road depth rather than the bow wave the vehicle actually sits in.

Scope

ESTABLISHED 20 coupled runs. All 17 that carry a determinism record are bit-reproducible; the 3 dry-start runs record none. Mesh containment 100.00 pct of a 2000-particle subsample. DxV bit-identical across a 2.1x mass range.

OPEN No reconstructed scene has entered a simulation. Vehicle velocity is zero throughout, a stationary-vehicle study, which is what AR&R itself measures. One hull, three masses, not three vehicles. Realized density above this project's stated band. Passthrough 7.3 to 15.9 pct. The zero out-of-bounds count is a post-clamp residual: 6,345 to 207,415 particle-frames were clipped back inside first. The 0.05 m onset threshold is this project's own, uncited.

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